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RESEARCH ARTICLE

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Cooperative surface-enhanced Raman spectroscopy enhancement in Au nanorod/SiO_2 nanoparticle solutions

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Abstract

Surface-enhanced Raman spectroscopy (SERS) signals in liquid state are significantly enhanced by utilizing cooperative interaction between metal surface plasmon and dielectric resonance. Raman signals from the diluted solutions are very weak even if they are amplified by SERS using Au nanorods. When SiO_2 nanoparticles are added together with Au nanorods, however, Raman intensity increased by three order comparing with that of system containing only Au nanorods. Finite-difference time-domain simulations show that SiO_2 nanoparticles exhibit dipolar electric resonance, which is strongly enhanced by interacting with the surface plasmon of Au nanorods. The size and concentration of SiO_2 nanoparticles are optimized to 354 nm in diameter and 4.5 vol%. Under the optimized condition, SERS intensity decrease with concentration of analyte (rhodamine 6G or crystal violet) is much slower for the system containing both Au nanorods and SiO_2 nanoparticles than that of the system containing only Au nanorods. The detection limit is 10^{-10} M for both aqueous rhodamine 6G and crystal violet solutions.

KEYWORDS

dielectric nanoparticles, liquid-state SERS, metal-dielectric coupling, SERS enhancement, volume exclusion

1 | INTRODUCTION

Raman spectroscopy has been a widespread analytical tool because of their capability of delivering rich chemical, structural, and compositional information as well as excellent specificity.[1–6] Although the Raman scattering signal is intrinsically very weak, it can be significantly enhanced by the so-called surface-enhanced Raman spectroscopy (SERS) technique.[7,8] SERS exploits the excitation of localized surface plasmon produced when light is irradiated on metal surfaces. In this case, metals are mainly in the form of nanoparticles or roughened surfaces.[9–12] The Raman scattering intensity is dependent on the strength of electric field ($I \approx |E|^4$), so the hot spots, the highly localized electric fields by surface plasmon, are responsible for the SERS enhancement. SERS signal also can be enhanced by chemical enhancement mechanism, which involves charge transfer between metal and molecules.[13–16] Due to the leaping developments of SERS-active materials and the analysis techniques, detection and even imaging of single molecules are now achievable.[17–24] Because SERS possesses strong distance dependence of $I \sim r^{-12}$, analytes for solid- or liquid-state SERS are demanded to aggregate close at the interparticle junctions to achieve high sensitivity.[25–29] For this reason, analytes are usually sandwiched between metal nanoparticle dimers using a variety of anchoring agents such as organic molecules, antibodies, proteins, DNAs

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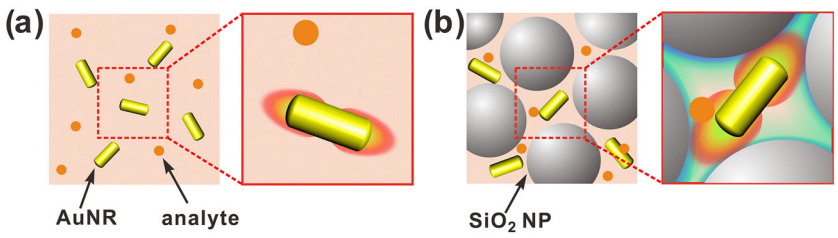
and polymers prior to SERS analysis.[30–32] For many of the practical cases where analytes are dissolved in solutions without anchoring to metal surface, or in situ reactions are monitored, however, SERS signals are relatively weak.

Herein, we report SERS enhancement by introducing dielectric SiO_2 nanoparticles (SiO_2 NPs) together with Au nanorods (AuNRs) in solution state (Scheme 1). Our approach utilizes (a) dielectric SiO_2 NPs as nanoantennas for enhancing surface plasmon response of AuNRs and (b) the large exclusion volume of SiO_2 NPs. As mentioned above, the common choice of nanoparticles for enhancing the interaction of light and molecules was limited to noble metals. However, dielectric particles also provide unique capability of manipulating electromagnetic waves at the nanometer scale. For example, dielectric particles have been used as microlenses,[33,34] whispering-galley mode resonators,[35–38] and photonic band gap materials.[39–41] Dielectric nanoparticle ensemble also can localize electric fields at confined space by multiple scattering of light.[42–46] In this case, unlike metal nanoparticles forming only electric dipoles, dielectric nanoparticles form both electric and magnetic dipole resonances due to low field loss in the particles.[47–49] When metal nanoparticles are suspended together with dielectric nanoparticles, it is expected that surface plasmon of metal nanoparticles and electric dipoles of dielectric nanoparticles are simultaneously generated upon exposure of light, and they can interact and create hybridized resonance mode by strong dipole–dipole coupling.[50–52] Same as the case of dipole–dipole coupling between metal nanoparticles, dielectric-metal dipole coupling leads to strong enhancement of electric field.[50–52] Moreover, it is also expected that large exclusion volume of dielectric nanoparticles confines analytes nearby metal nanoparticles and increase their local concentration. In this context, SERS enhancement in solution by addition of dielectric/metal nanoparticle ensemble has been investigated using model system consisting of SiO_2 NPs as dielectric nanoparticles, AuNRs as metal nanorods, and rhodamine 6G (R6G) and crystal violet (CV) as analyte molecules. Our results show that when SiO_2 NPs were added along with AuNRs, SERS signals and the detection limit were dramatically increased by almost 10^3 times comparing with the cases containing only AuNRs.

2 | RESULTS AND DISCUSSION

As illustrated in Scheme 1, cooperative SERS enhancement occurs in the colloidal mixture of AuNRs and SiO_2 NPs. To prepare the solution mixture, AuNRs and SiO_2 NPs were first synthesized separately and then mixed together with a certain ratio. AuNRs were synthesized following the procedure reported elsewhere.[53] Briefly, gold nanoparticle seeds for the synthesis of AuNRs were first prepared from $HAuCl_4$ (5.0 ml, 0.5 mM) by adding $NaBH_4$ (0.6 ml, 0.01 M) in the presence of CTAB (5.0 ml, 0.2 M). Once $NaBH_4$ was added, the solution was turned to brownish yellow, indicating the formation of gold nanoparticles. The growth solution was separately prepared by mixing CTAB (5.0 ml, 0.2 M), $AgNO_3$ (250 μl, 4 mM), $HAuCl_4$ (5.0 ml, 1.0 mM), and ascorbic acid (70 μl, 78.8 mM) at 25°C. AuNRs were grown by adding 12 μl of the presynthesized seed solution to the growth solution with vigorous stirring, and then the solution was aged for 2 hr at 30°C. The synthesized AuNRs were characterized by using TEM and UV–VIS (Figure S1). The average diameter and length of the synthesized AuNRs was 10 ± 9 and 41 ± 8 nm, respectively. The transversal surface plasmon peak was observed at λ = 510 nm, and the longitudinal one was at λ = 785 nm. SiO_2 NPs were synthesized by using the conventional Stöber method.[54] Briefly, ethanol (21 ml), DI-water (7 ml), and ammonium hydroxide (1.4 ml) were first mixed, and then TEOS (2 mL) was added to the solution dropwise (0.4 ml/hr) with vigorous stirring at room temperature. After 24 hr of reaction at room temperature, the silica particles were isolated by centrifugation at 7,077 RCF for 20 min and rinsed several times to eliminate residues of reagents. The size of SiO_2 NPs was controlled by changing the concentration of TEOS or ammonium hydroxide (Figure S2). The isolated AuNRs and SiO_2 NPs were then mixed together with probe molecules, R6G. Raman spectral collections were carried out with the wide area illumination scheme (Scheme S1). The



SCHEME 1 Schematic illustration of (a) SERS enhancement by only AuNR and (b) cooperative SERS enhancement by AuNR and SiO_2 NP ensemble. Cooperative enhancement occurs by coupling of metal surface plasmon with dielectric dipoles [Colour figure can be viewed at wileyonlinelibrary.com]

illumination source was a 785-nm laser with a 6-mm beam size operated at 400 mW. Due to its much larger illumination area (28.3 mm^2) than conventional dispersive and FT-Raman (spot size = 50–500 μm), wide area illumination Raman provides highly reproducible and reliable SERS signals.[55,56] The spectral band of the illumination laser was matched with the longitudinal plasmon resonance of AuNRs.

As shown in Figure 1, the SERS signals were significantly enhanced by the addition of SiO_2 NPs. R6G dissolved in water with a concentration of 10^{-3} M exhibited five characteristic Raman peaks at 1,179, 1,310, 1,350, 1,362, and 1,510 cm^{-1} (Figure 1b). However, those peaks were barely distinguishable when the concentration was lowered to 10^{-8} M (Figure 1c). At such low concentration (10^{-8} M), R6G peaks were not observed even after addition of AuNRs (10^{-10} M) for SERS enhancement (Figure 1d). The four peaks shown in Figure 1d (1,130, 1,235, 1,400, and 1,450 cm^{-1}) correspond to those of AuNRs, and they are originated from CTAB molecules that were used as surfactants for the synthesis of AuNRs.[57] In contrast, the Raman peaks of R6G were strongly enhanced when SiO_2 NPs were added to the aqueous mixture of R6G (10^{-8} M) and AuNRs (10^{-10} M) (Figure 1e). As shown in Figure 1e, all peaks of R6G were clearly observable. Interestingly, most of AuNR peaks were suppressed in the mixture of AuNRs and SiO_2 NPs. Although the same concentration of AuNRs (10^{-10} M) was used for both samples shown in Figure 1d,e, the first three peaks of AuNRs (1,130, 1,235, and 1,400 cm^{-1}) were almost disappeared and, oppositely, the fourth peak at 1,450 cm^{-1} was enhanced. In the aspect of Raman analysis, fewer or lower background peak is desirable. It is noteworthy that our SERS enhancement by SiO_2 NPs is different from the case of enhancement by the photonic crystals.[39,58] Although regularly assembled 2D or 3D dielectric photonic crystals are known to strongly enhance Raman scattering, the concentration of SiO_2 NPs used in our experiment (4.5 vol%) was too low to form photonic structures. SiO_2 NPs at such low concentration are randomly dispersed in water.

SERS enhancement by SiO_2 NPs was highly dependent on the concentration and the size of SiO_2 NPs. To test the effects of SiO_2 NPs, three different size of SiO_2 NPs (d = 101, 354, and 637 nm) were added to the mixture of R6G (10^{-8} M) and AuNRs (10^{-10} M). As shown in Figure 2, the Raman intensity of R6G sharply increased upon addition of small amounts of SiO_2 NPs and showed the maximum enhancement at 4.5 vol% for 354- and 637-nm size or at 12.5 vol% for 101-nm size of SiO_2 NPs. SERS enhancement decreased again when the concentration of SiO_2 NPs was higher than the optimal concentrations. We attribute this behavior to the increase of light extinction by SiO_2 NPs at high concentrations. Due to the increasing turbidity with the concentration of SiO_2 NPs, less probe light can reach inside of sample. The size of SiO_2 NPs also significantly affects the SERS enhancement. As shown in Figure 2, 354-nm size SiO_2 NPs showed much stronger

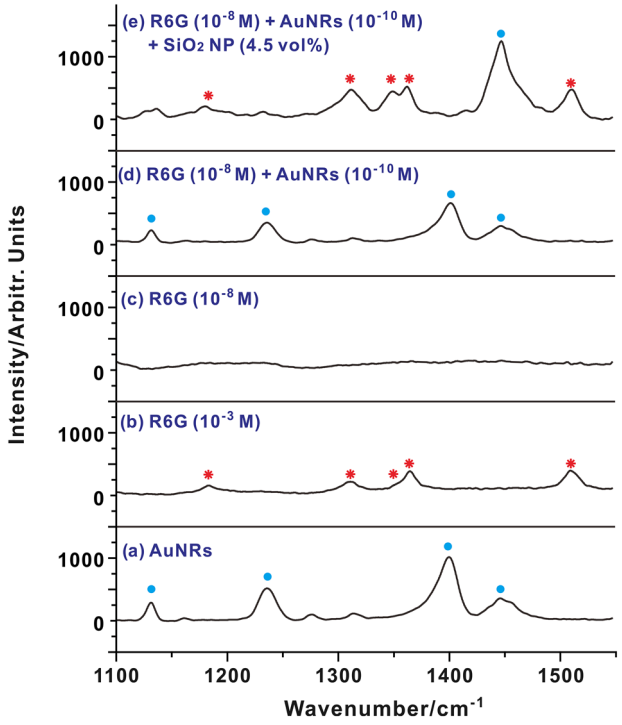


FIGURE 1 Raman spectra of (a) AuNRs dispersion in water, (b) aqueous R6G (10^{-3} M), (c) aqueous R6G (10^{-8} M), (d) aqueous mixture of R6G (10^{-8} M) and AuNRs (10^{-10} M), and (e) aqueous mixture of R6G (10^{-8} M), AuNRs (10^{-10} M), and SiO_2 NPs (4.5 vol%) [Colour figure can be viewed at wileyonlinelibrary.com]

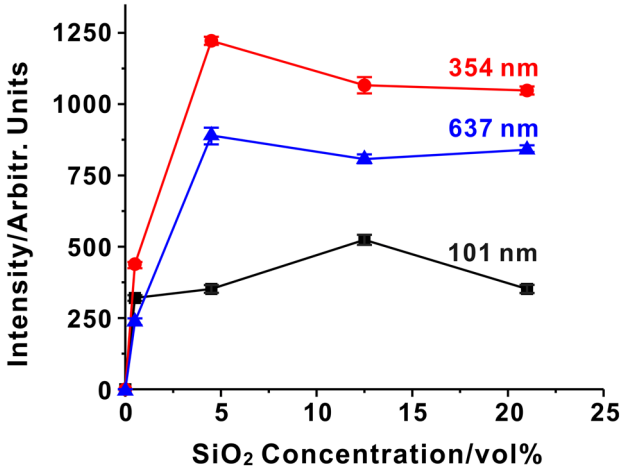


FIGURE 2 Raman scattering intensity changes of R6G at 1,510 cm^{-1} as functions of the size and the concentration of SiO_2 NPs. The concentration of R6G and AuNRs was 10^{-8} M and 10^{-10} M, respectively [Colour figure can be viewed at wileyonlinelibrary.com]

SERS enhancement than 101- or 637-nm size of SiO_2 NPs. We attribute this to the size and refractive index dependent scattering efficiency. For the given refractive index of SiO_2 NPs ($n = 1.54$), the maximum of forward to backward scattering efficiency (Q_{FS}/Q_{BS}) occurs where the size parameter ($q = 2\pi R/\lambda$) is ranged around $q \approx 1.5$.[47,48] The calculated q values for SiO_2 NPs are $q = 0.4$, 1.4, and 2.4 for 101, 354, and 637 nm, respectively. Hence, 354 nm of SiO_2 NPs that have the closest q value to 1.5 can have the most efficient forward scattering. It is expected that the more forward scattering can induce more dipole–dipole interactions between SiO_2 NPs and AuNRs and accordingly the better SERS enhancement. Based on our experiments, the optimal condition for SiO_2 NPs was determined to 4.5 vol% of 354-nm size SiO_2 NPs.

SERS enhancement by the addition of SiO_2 NPs was well consistent with the theoretical simulations. Figure 3 shows the electric field distribution profiles calculated by finite-difference time-domain simulation. In this simulation, the dimension and the concentration of AuNRs and SiO_2 NPs were matched with those of actual experimental conditions, and also, monochromatic light ($\lambda_{peak} = 785$ nm) is irradiated along the z-axis. The snapshots are representative images showing the highest electric field strength for each system. Simulation results show that the electric field enhancement is cooperative. The electric field strength is significantly enhanced in the mixture of AuNRs/SiO_2 NPs (Figure 3c) comparing with the cases where AuNRs (Figure 3a) or SiO_2 NPs (Figure 3b) were individually present. In this case, SiO_2 NPs showed strong dipolar electric field distribution. Dipolar electric distribution is more apparent for single SiO_2 NP simulation (Figure S3). As shown in Figure S3, SiO_2 NP exhibited an apparent electric dipole, which was strongly enhanced by AuNRs especially nearby the

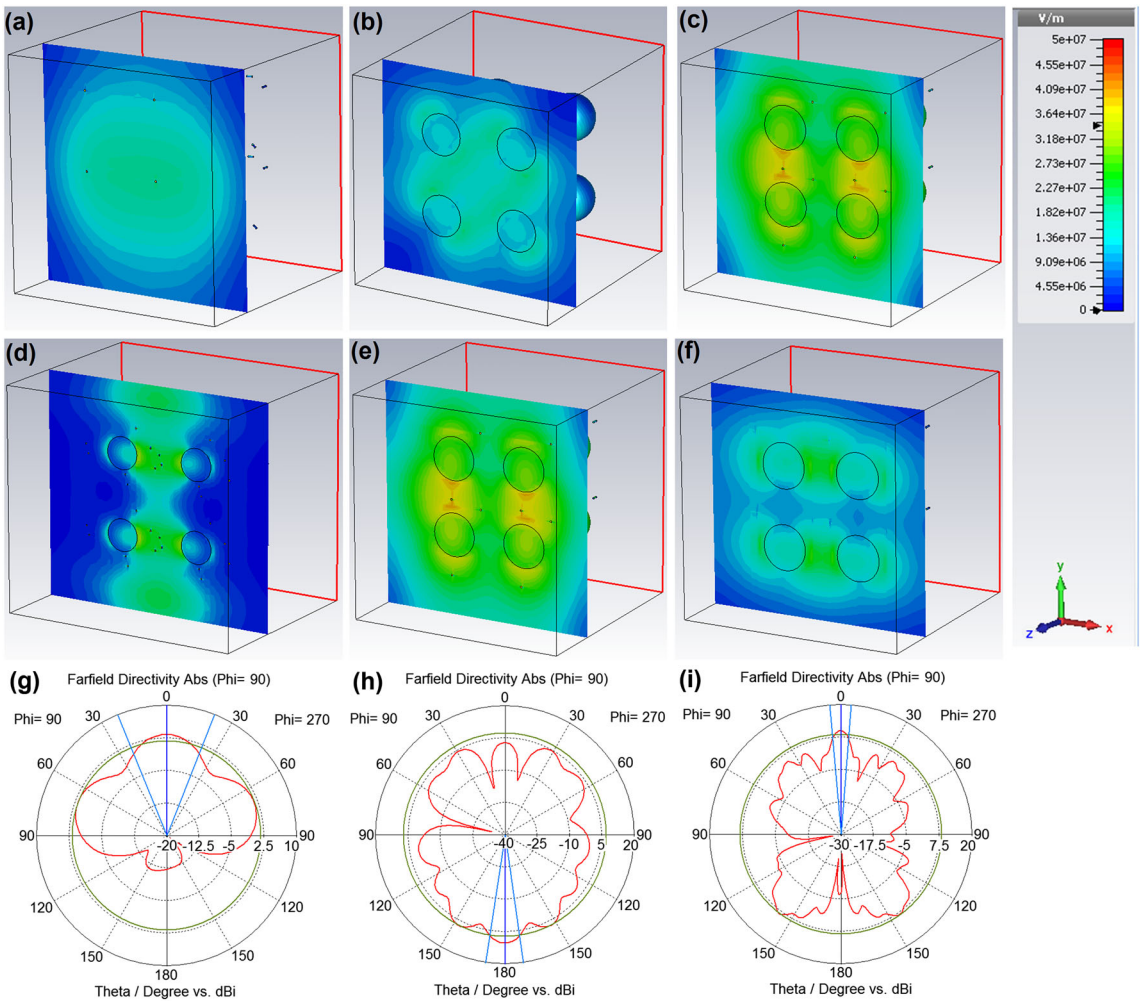


FIGURE 3 FDTD simulation results. (a–c) Electric filed distribution of (a) pure AuNRs ($d = 10$ nm $L = 41$ nm, $c = 10^{-10}$ M), (b) pure SiO_2 NPs ($d = 354$ nm, $c = 4.5$ vol%), and (c) mixture of AuNRs ($d = 10$ nm $L = 41$ nm, $c = 10^{-10}$ M) and SiO_2 NPs ($d = 354$ nm, $c = 4.5$ vol%). (d–f) Electric field distribution of AuNRs/SiO_2 NPs solutions with different concentration of SiO_2 NPs ($d = 354$ nm): (d) $c = 2$ vol%, (e) $c = 4.5$ vol%, and (f) $c = 6.3$ vol%. (g–i) Radiation pattern of AuNRs/SiO_2 NPs solutions with different size of SiO_2 NPs ($c = 4.5$ vol%): (g) $d = 101$ nm, (h) $d = 354$ nm, and (i) $d = 637$ nm [Colour figure can be viewed at wileyonlinelibrary.com]

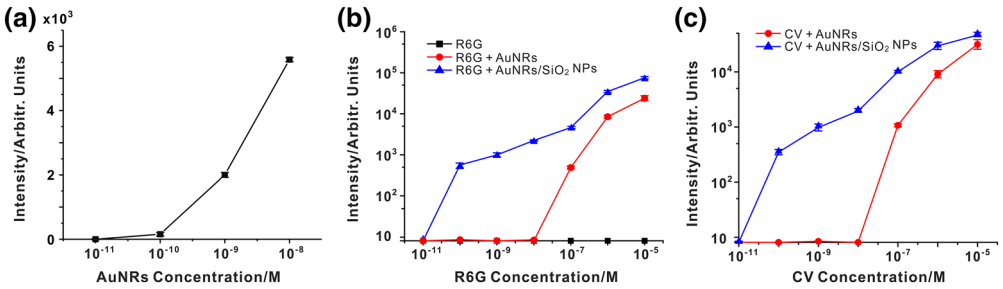


FIGURE 4 (a) SERS intensity change of R6G (1,510 cm^{-1}) as a function of AuNRs. (b) Detection limit changes of R6G (1,510 cm^{-1}) by addition of SiO_2 NPs. Addition of both AuNRs/SiO_2 NPs increased the detection limit by almost three order of the case containing only AuNRs. Pure R6G was undetectable below $c_{R6G} = 10^{-4}$M. (c) Detection limit changes of CV (1,171 cm^{-1}) by addition of SiO_2 NPs. The same as R6G, the detection limit of CV increased by three order by addition of SiO_2 NPs [Colour figure can be viewed at wileyonlinelibrary.com]

surface of SiO_2 NP. It is notable that the dipolar electric fields were enhanced between two SiO_2 NPs pairs even they were apart several hundred nanometers (Figure 3c). These results suggest that electric dipoles of SiO_2 NPs interact with surface plasmon of AuNRs and SiO_2 NP themselves.[51] The surface plasmon-dielectric dipole interaction between AuNRs and SiO_2 NPs was further supported by the red-shift of surface plasmon peaks of AuNRs when SiO_2 NPs were added to AuNRs dispersion (Figure S4). In this case, the red-shift of both transversal and longitudinal peaks suggests that the surface plasmon peak shift was not simply by aggregation of nanorods. Aggregation of nanorods leads to the strong blue-shift of the longitudinal peak because of the decreased aspect ratio. In this case, due to the red-shift, the resonance peak of AuNRs was slightly deviated from the excitation wavelength. It is expected that the detection limit can be further improved by controlling the geometry of AuNRs matching the resonance peak with the excitation wavelength. SERS changes by the concentration of SiO_2 NPs also agree well with the experimental results (Figure 3d–f). Same as our experimental results, the strongest electric field was observed at 4.5 vol% of SiO_2 NPs (Figure 3d–f). The size effect of SiO_2 NPs was also well demonstrated by simulation (Figure 3g–i). The radiation patterns showed that 354-nm SiO_2 NPs exhibited strong forward scattering, whereas back scattering was dominant for 101- and 637-nm SiO_2 NPs. These results are well consistent with our previous explanation based on the size parameter.

SERS intensity increased with the concentration of AuNRs (Figure 4a). The concentration of AuNRs was optimized to $c_{AuNRs} = 10^{-8}$ M. Although the detection limit increased as increasing the concentration of AuNRs, AuNRs were not stable above $c_{AuNRs} = 10^{-7}$ M. To verify the detection limit, SERS measurements have been done at the optimized conditions of AuNRs (10^{-8} M) and SiO_2 NPs (354 nm, 4.5 vol%). The intensity of R6G Raman signal at 1,510 cm^{-1} was measured as a function of the R6G concentration for each system, R6G, R6G/AuNRs, and R6G/AuNRs/SiO_2 NPs, respectively (Figure 4b). Raman scattering of aqueous R6G was barely detectable below $c_{R6G} = 10^{-4}$ M. When AuNRs were added, the detection limit was improved to $c_{R6G} = 10^{-7}$ M. When both AuNRs and SiO_2 NPs were added, the detection limit was further improved to $c_{R6G} = 10^{-10}$ M. In this case, it is notable that the slope of the intensity change graph became slower for the system containing both AuNRs and SiO_2 NPs than that of the case containing only AuNRs. Hence, the Raman intensity difference between the two systems became more prominent at a low concentration regime ($c_{R6G} \leq 10^{-7}$ M), whereas the difference was relatively small at a high concentration regime ($c_{R6G} \geq 10^{-6}$ M). Such effective SERS enhancement at low concentration of R6G was also partially attributed to the large volume exclusion of SiO_2 NPs. The similar SERS enhancements were observed for other dyes. As shown in Figure 4c, the SERS signals of crystal violet (CV) were also significantly enhanced when both AuNRs and SiO_2 NPs were added to CV solutions, and the intensity changes with the concentration of CV were almost the same as the case of R6G.

3 | CONCLUSION

In summary, we have successfully demonstrated SERS enhancement in liquid state by utilizing cooperative interaction of dielectric resonance of SiO_2 NPs with surface plasmon of AuNRs. Comparing with the system containing only AuNRs, the electric field strength was significantly increased when both AuNRs and SiO_2 NPs were added together, which leads to strong enhancement of SERS signals. In this case, electric dipolar resonance of SiO_2 NPs was strikingly enhanced by coupling with the surface plasmon of AuNRs. SERS enhancement by addition of SiO_2 NPs was highly

dependent on the concentration and the size of SiO_2 NPs, and they were optimized to 4.5 vol% and 354 nm in diameter, respectively. Under the optimized condition, SERS enhancement was demonstrated for analytes, R6G, and CV. For both cases, the detection limit and SERS intensity of system containing AuNRs/SiO_2 NPs were increased by three order comparing with the system containing only AuNRs. We anticipate that our system is highly versatile to apply other SERS systems to enhance SERS signals.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article.

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